Beautiful Dua from Quran

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي

وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

Fall

(2024)

Q. What is the type of hybridization in Nickel in Ni $(CO)_4$ a and $[Ni(CN)_4]^{-2}$

Haq Nawaz Bhatti Page No. 157 onward

Sanaullah Page No. 206 and 208

Q. Metals do not form ionic or covalent bonds?

Metals, in their pure or elemental form, do not form ionic or covalent bonds. Instead, they form **metallic bonds**, which are a special type of chemical bond. In metallic bonding, each metal atom releases its outer (valence) electrons, and these electrons move freely among all the atoms. This creates a "sea of electrons" that surrounds positively charged metal ions (atoms that have lost electrons).

This unique bonding structure is responsible for the characteristic properties of metals. The freely moving electrons allow metals to conduct electricity and heat. Also, this bonding lets metal layers slide over each other without breaking, which makes metals **malleable** (easily shaped or bent) and **ductile** (can be drawn into wires).

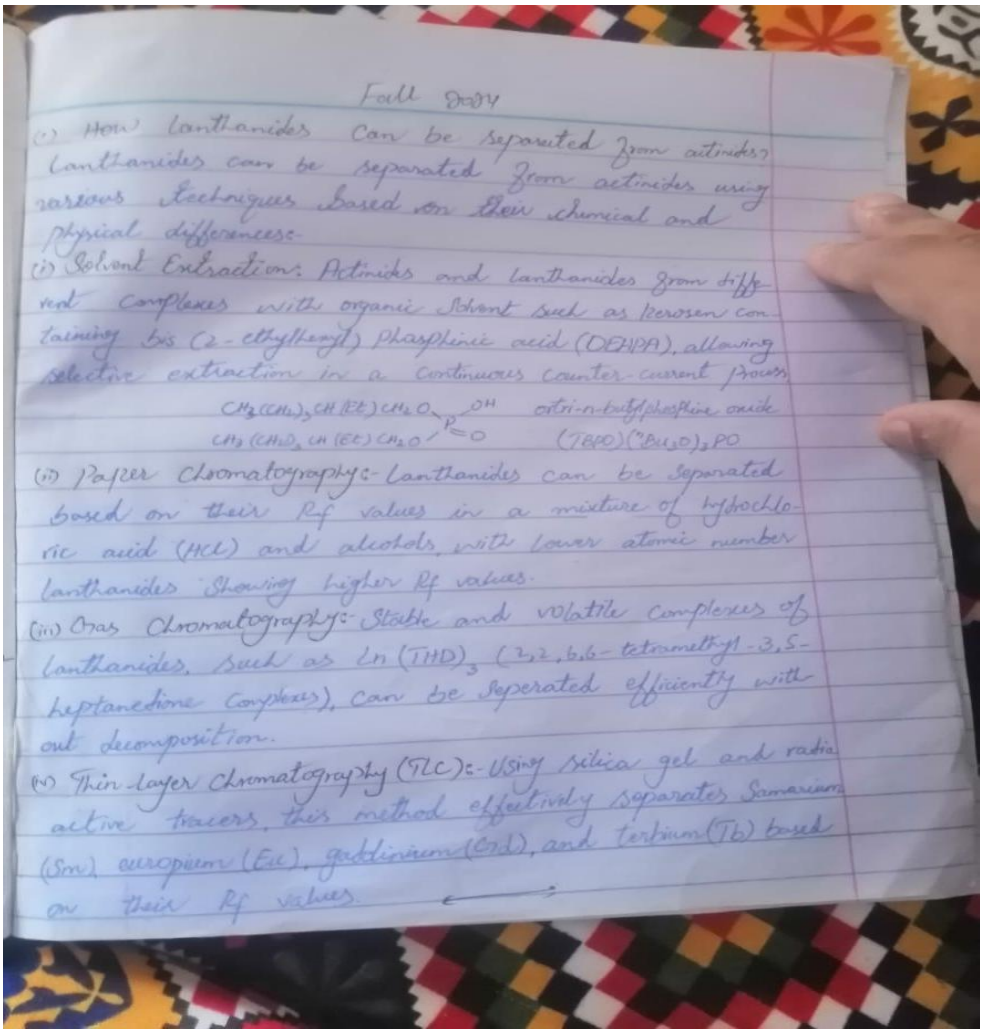
Example of Metallic Bonding (in Pure Sodium – Na):

In pure sodium metal, each sodium (Na) atom **loses one valence electron**, which becomes part of the **"sea of electrons."** The remaining atom becomes a **positive ion (Na^{+})**.

$$\mathbf{Na \rightarrow Na^{+} + e^{-}}$$

These **Na^{+} ions** are arranged in a regular pattern (lattice), and the **free-moving electrons** hold the entire structure together. This is why sodium metal can **conduct electricity** and is **malleable** and **ductile**.

Q.How lanthanides can be separated from actinides?



Fall 2024

(i) How lanthanides can be separated from actinides?

Lanthanides can be separated from actinides using various techniques based on their chemical and physical differences.

(i) Solvent Extraction: Actinides and lanthanides from different complexes with organic solvent such as kerosen containing bis (2-ethylhexyl) phosphinic acid (DEHPA), allowing selective extraction in a continuous counter-current process.

$CH_3(CH_2)_3CH(Et)CH_2O$, $CH_3(CH_2)_3CH(Et)CH_2O$ > P(OH)=O — ortri-n-butylphosphine oxide (TBPO) $(^nBu_3O)_3PO$

(ii) Paper Chromatography:- Lanthanides can be separated based on their Rf values in a mixture of hydrochloric acid (HCl) and alcohols, with lower atomic number lanthanides showing higher Rf values.

(iii) Gas Chromatography:- Stable and volatile complexes of lanthanides, such as $Ln(THD)_3$ (2,2,6,6-tetramethyl-3,5-heptanedione complexes), can be seperated efficiently without decomposition.

(iv) Thin-layer Chromatography (TLC):- Using silica gel and radioactive tracers, this method effectively separates Samarium (Sm), europium (Eu), gadolinium (Gd), and terbium (Tb) based on their Rf values.

Q. Neither La^{3+} nor Lu^{3+} ion show any colour?why

The **La^{3+}** and **Lu^{3+}** ions are colorless because they have no electrons in their 4f orbitals, resulting in no f-f electron transitions that could absorb visible light and produce color.

- **La^{3} (Lanthanum):** Lanthanum's electron configuration is [Xe] $5d^1$ $6s^2$. When it forms La^3, it loses the $5d^1$ and $6s^2$ electrons, leaving an electron configuration of [Xe], which has a completely empty 4f orbital ($4f^0$). With no electrons in the 4f orbitals, there are no f-f transitions possible, and thus, La^{3+} does not absorb visible light, appearing colorless.
- **Lu^{3} (Lutetium):** Lutetium's electron configuration is [Xe] $4f^{14}$ $5d^1$ $6s^2$. Upon forming Lu^3, it loses the $5d^1$ and $6s^2$ electrons, resulting in [Xe] $4f^{14}$. Here, the 4f orbitals are completely filled. Since f-f transitions require partially filled orbitals to allow electron movement between energy levels, a fully occupied $4f^{14}$ configuration cannot undergo such transitions, making Lu^3 colorless as well.

In lanthanide ions, color typically arises from f-f electronic transitions, which occur when 4f orbitals are partially filled (e.g., in ions like Ce^3 or Nd^3). Since La^3 has an empty 4f orbital and Lu^3 has a fully filled 4f orbital, neither can produce these transitions, resulting in no absorption in the visible spectrum and, consequently, no color.

Q.What are the toxic effects of actinides?

Actinides are a group of 15 radioactive metallic elements from Actinium (Ac) to Lawrencium (Lr) with atomic numbers 89 to 103. These elements are both radioactive and chemically toxic. When they enter the human body through inhalation, ingestion, or wounds, they can cause severe health issues due to radiation and heavy metal toxicity.

1. Radiological Toxicity:

Actinides emit alpha, beta, and gamma radiation. Alpha particles are highly damaging when emitted inside the body and can cause cell death and DNA mutations.

Example: Inhalation of Plutonium-239 (Pu-239) particles leads to lung cancer because alpha particles damage lung tissues.

Example: Americium-241 (Am-241) emits both alpha and gamma rays and accumulates in the bone marrow, increasing the risk of leukemia.

2. Chemical Toxicity:

As heavy metals, actinides disrupt biological functions by mimicking essential metals like calcium and iron, leading to organ damage.

Example: Uranium (U-238) gets deposited in bones and kidneys, causing nephrotoxicity (kidney damage) and bone marrow suppression.

Example: Neptunium-237 (Np-237) affects liver and immune function, especially in soluble chemical forms.

3. Environmental and Exposure Pathways:

Actinides enter the body through exposure to nuclear materials, waste, or weapons. They persist in the environment and can contaminate air, water, and soil.

Example: Depleted Uranium (DU) used in ammunition produces fine dust particles that cause genetic mutations and lung problems when inhaled.

Example: Curium (Cm), Californium (Cf), and Berkelium (Bk) are synthetic actinides that can spread through groundwater and pose long-term radiological risks.

Q.What is used as coolant in nuclear reactors?

Haq Nawaz Bhatti Page No. 113

Q.Which element is used as an energy source in heart pacemakers?

A **heart pacemaker** is a small electronic medical device that is **implanted inside the chest** to help control **abnormal heart rhythms** (arrhythmias). It sends **electrical impulses** to the heart muscles to ensure that the heart beats at a **normal and steady rate**. Pacemakers are often used in patients who suffer from **bradycardia** (slow heart rate) or **heart block**, where the natural electrical system of the heart is damaged or weakened. The device helps improve **blood circulation**, prevent **fatigue**, and reduce the risk of **heart failure or cardiac arrest**.

A special radioactive element called **Plutonium-238 (Pu-238)** was used as the **energy source** in some heart pacemakers. Plutonium-238 is a **radioactive isotope** that emits **alpha radiation**, which is not highly penetrating but releases a steady amount of **heat energy**. This heat was converted into **electricity** using a **thermoelectric generator** inside the pacemaker, ensuring a **long-lasting power supply**. Due to its **long half-life of 87.7 years**, Pu-238-powered pacemakers could operate for several decades without needing a battery replacement — making them ideal for patients who were not good candidates for repeat surgeries.

However, due to concerns about **radiation exposure**, **cost**, and the availability of safer alternatives, the use of Pu-238 has been largely discontinued in modern medical practice. Today, most pacemakers use **lithium-based batteries**, which are **compact, safe, and last between 8 to 15 years**. Although Pu-238-powered pacemakers are now rare, they were a remarkable use of **nuclear technology in life-saving medical applications**.

Q.What is misch metal? What are its uses?

Read Page No 465 Haq Nawaz Bhatti

Mischmetal is an alloy made from a mixture of rare-earth elements. It is obtained from minerals like monazite and bastnasite and is silvery and brittle in appearance. It is non-radioactive and used in various industrial applications.

Typical composition includes:

- 50–55% Cerium (Ce)
- 20–25% Lanthanum (La)

- 15–18% Neodymium (Nd)
- 4–5% Praseodymium (Pr)
- ~5% Other lanthanides + small amounts of Iron (Fe)

Uses of Mischmetal:

1. Flint for Lighters:
Mischmetal is used in making ferrocerium for lighter flints. Cerium in the alloy produces sparks when scraped. It is used in cigarette lighters, welding torches, and fire starters.

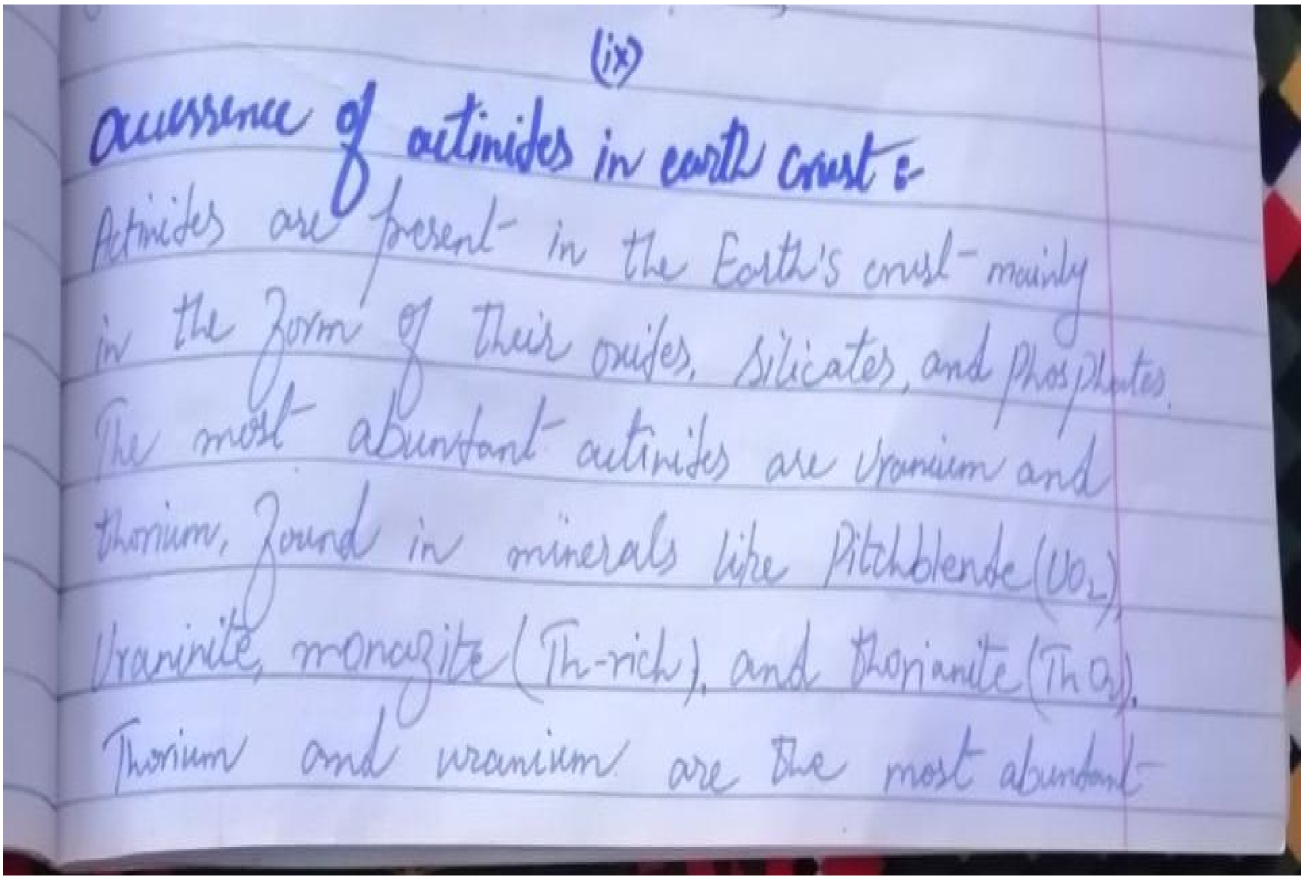
2. Metallurgical Deoxidizer:
In steel and iron industries, mischmetal is used to remove oxygen and sulfur. This improves metal quality and strength. It is especially useful in casting and refining processes.

3. Nickel-Metal Hydride (Ni-MH) Batteries:
Mischmetal is a component in hydrogen-storing alloys used in batteries. These batteries are found in hybrid cars and electronics. It provides better energy storage and recharge cycles.

What is hybridization in AB9 molecule?

Read Page No 162 Haq Nawaz Bhatti

Q.Briefly write the occurrence of actinides in earth crust?


(ix)
occurrence of actinides in earth crust :-
Actinides are present in the Earth's crust mainly in the form of their oxides, silicates, and phosphates. The most abundant actinides are Uranium and thorium, found in minerals like Pitchblende (UO_2), Uraninite, monazite (Th-rich), and thorianite (ThO_2). Thorium and uranium are the most abundant

actinides in nature with the respective mass concentrations of 1.6×10^{-3} % and 4×10^{-4} %.

*Uranium occurs in various ores such as Carnotite ($K_2(UO_2)_2(VO_4)_2 \cdot 3H_2O$) and autunite ($Ca(UO_2)_2(PO_4)_2 \cdot 10\text{-}12H_2O$). It is widely mined in Canada, Kazakhstan, Australia, Namibia, and Russia.

*Thorium is mainly found in monazite and thorite. Major deposits exist in India, the USA, Canada, and Brazil.

* Other actinides like Plutonium, neptunium, americium, and curium are found in trace amounts, mostly as decay products of uranium. Plutonium is primarily produced synthetically in nuclear reactors.

*Actinium is very rare, occurring in uranium-bearing minerals due to the decay of ^{235}U.

These actinides are crucial for nuclear energy production and scientific research.

Q.What is the effective nuclear charge experience by a valence p- electron in boron?

Effective Nuclear Charge (Z_eff) is the net positive charge experienced by an electron in a multi-electron atom. Due to the shielding effect of inner electrons, the outer (valence) electrons do not feel the full attractive force of the nucleus.

Formula:

$$Z_{eff} = Z - S$$

Where:

- Z = Atomic number of the element
- S = Shielding or screening constant (estimated using Slater's Rules)

For Boron (B):

- ❖ Atomic number, Z=5
- ❖ Electron configuration = $1s^2\ 2s^2\ 2p^1$
- ❖ We are calculating $\mathbf{Z_{eff}}$ for a 2p electron (valence p-electron).

Now apply Slater's Rules to find S:

1. Group electrons:

($1s^2$)
($2s^2$, $2p^1$) ← group of interest

2. Apply Slater's Rules for 2p electron:

- ❖ Electrons in the same group (2s and 2p): each contributes 0.35
  → 2 electrons × 0.35 = 0.70
- ❖ Electrons in the n=1 shell ($1s^2$): each contributes 0.85
  → 2 electrons × 0.85 = 1.70
- ❖ Total S = 0.70 + 1.70 = 2.40

$$\mathbf{Z_{eff} = Z - S = 5 - 2.40 = 2.60}$$

<u>Read page number 8 Haq nawaz bhatti</u>

Q.Explain the magnetic behavior of O, molecule by comparing its VBT and MOT pictures?

The magnetic behavior of the oxygen molecule **(O_2)** can be explained by comparing the Valence Bond Theory (VBT) and Molecular Orbital Theory (MOT), which differ notably in their treatment of electron pairing and magnetism.

✅ Valence Bond Theory (VBT) Picture of O_2

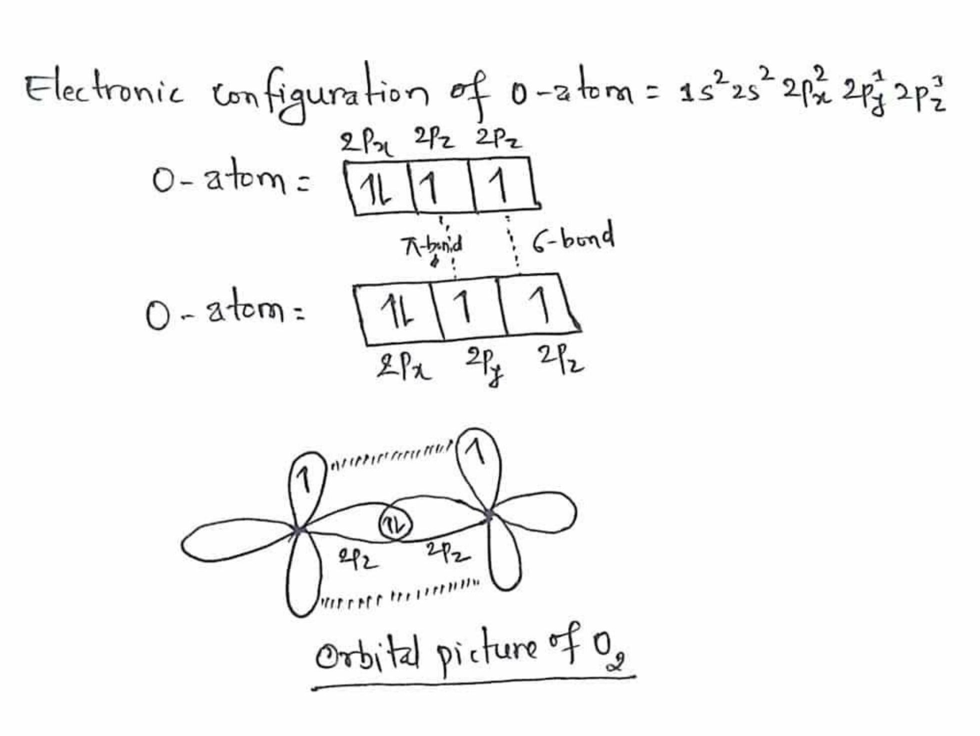
Each oxygen atom has the electronic configuration $1s^2\ 2s^2\ 2p^4$.
VBT says two oxygen atoms form a double bond:

- ❖ One sigma (σ) bond (head-on overlap of orbitals)
- ❖ One pi (π) bond (side-by-side overlap of p orbitals)

Electron Pairing:
VBT assumes all electrons involved in bonding are paired.
Bond Order:
Bond order = 2 (one σ + one π bond)

Magnetic Behavior:
Since all electrons are paired, VBT predicts O_2 to be diamagnetic (not attracted to magnetic fields).

Limitation:
This contradicts experimental observations. Oxygen is actually paramagnetic, which VBT fails to explain.



Molecular Orbital Theory (MOT) Picture of O_2

MOT treats electrons as occupying molecular orbitals that are spread over the entire molecule, not localized between atoms.

Total Electrons in O_2 :
8 electrons from each oxygen = 16 electrons

- **Molecular Orbital Filling Order:**

$$\sigma(1s)^2\,\sigma^*(1s)^2\,\sigma(2s)^2\,\sigma^*(2s)^2\,\sigma(2p_z)^2\,\pi(2p_x)^2 = \pi(2p_y)^2\,\pi^*(2p_x)^1\,\pi^*(2p_y)^1$$

- **Bond Order Calculation:**

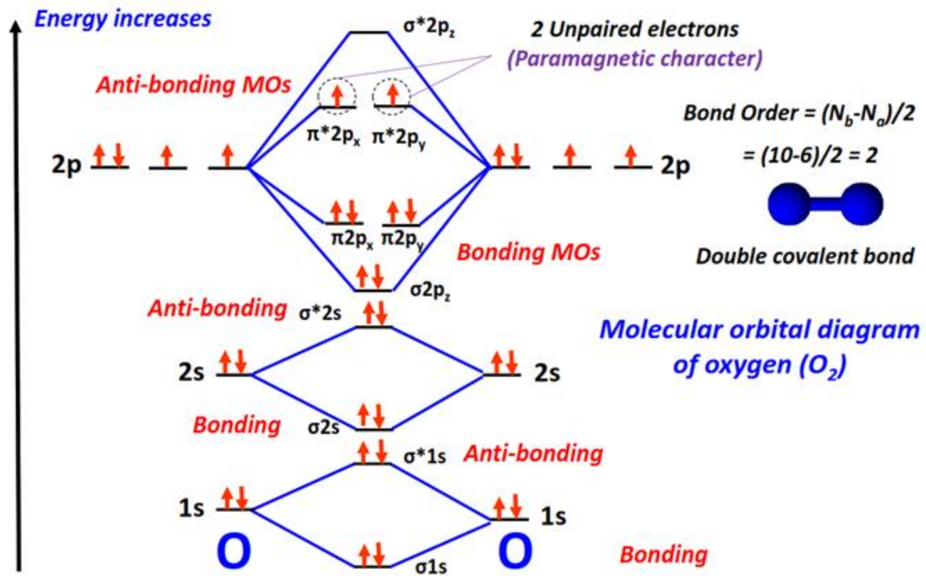
$$\text{Bond order} = \frac{N_b - N_a}{2} = \frac{10-6}{2} = 2$$

(Where N_b = bonding electrons, N_a = antibonding electrons)

Electron Pairing:
Two electrons are unpaired in the antibonding π^*2p orbitals.

Magnetic Behavior:
Because of these two unpaired electrons, MOT correctly predicts that O_2 is paramagnetic (attracted to magnetic fields).

Experimental Evidence:
Liquid oxygen is suspended between the poles of a magnet—clear proof of paramagnetism, explained only by MOT.



Q. Give two applications of metal complexes in biological systems?

1. Oxygen Transport – Hemoglobin (Iron Complex)

One important application of metal complexes in biological systems is **oxygen transport** in the human body. The protein **hemoglobin**, found in red blood cells, contains an **iron(II) (Fe^{2+}) complex** at the center of a **heme group** (a porphyrin ligand). This metal complex binds oxygen molecules reversibly. The reaction involved can be represented as:

$$\mathbf{Hb\text{-}Fe^{2+} + O_2 \rightleftharpoons Hb\text{-}Fe^{2+}\text{-}O_2}$$

In this reaction, the **Fe^{2+}** ion binds with O_2 to form **oxyhemoglobin**. This allows oxygen to be transported from the lungs to body tissues, where oxygen is released for cellular use.

2. Photosynthesis – Chlorophyll (Magnesium Complex)

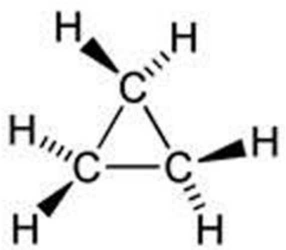
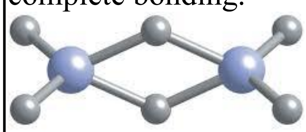
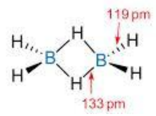
Another important application is in photosynthesis, where the metal complex chlorophyll plays a central role. Chlorophyll is a green pigment in plants that contains magnesium (Mg^{2+}) at the center of a porphyrin-like ring. It captures solar energy to convert carbon dioxide and water into glucose and oxygen.

The overall reaction is:

$$6CO_2 + 6H_2O \xrightarrow{\text{chlorophyll, sunlight}} C_6H_{12}O_6 + 6O_2$$

Here, the **Mg^{2+}** ion helps in the absorption of light energy, making photosynthesis possible. This process is vital for producing food and releasing oxygen into the atmosphere.

Q.What is the difference between bent bond and bridge bond?

Feature	**Bent Bond**	**Bridge Bond**
Definition	A bent bond is a type of covalent bond formed when two atomic orbitals overlap at an angle rather than in a straight line, resulting in a curved or off-axis bond.	A bridge bond is a type of covalent bond where one atom (usually hydrogen) forms a bond with two other atoms simultaneously, creating a three-center bond.
Shape	The shape of a bent bond is curved or banana-like due to the angular overlap of orbitals, which deviates from the usual straight bond geometry.	In a bridge bond, the bridging atom is located between two atoms, giving the appearance of a bridge connecting them.
Example	Bent bonds are found in molecules like **cyclopropane**, where the carbon-carbon bonds are bent due to ring strain and reduced bond angles. 	Bridge bonds are observed in molecules like **diborane (B_2H_6),** where hydrogen atoms bridge two boron atoms to complete bonding.  
Bond Type	A bent bond is a **2-center 2-electron (2c–2e)** bond, meaning two atoms share two electrons in a distorted bonding arrangement.	A bridge bond is a **3-center 2-electron (3c–2e)** bond, in which three atoms share two electrons, common in electron-deficient compounds.
Electron Sharing	In bent bonds, electrons are shared between two atoms only, but the overlap is not linear due to molecular geometry or strain.	In bridge bonds, the electrons are shared among three atoms, allowing bonding in molecules that do not have enough electrons for standard bonding.
Cause	Bent bonds are typically caused by orbital strain or forced small bond angles that prevent the orbitals from aligning in a straight line.	Bridge bonds arise mainly due to electron deficiency, especially in elements like boron and aluminum, where normal 2c–2e bonding is insufficient.

Q.Give any method of preparation of metal complexes?

A number of preparative techniques have been used to prepare metal carbonyls, and four general ones will be discussed here.

1. Direct Synthesis

Carbon monoxide reacts with several finally divided metals to give metal carbonyls. Nickl, iron and cobalt carbonyls can be prepared in this way.

$$Ni + 4CO \longrightarrow Ni(CO)_4$$

$$Fe + 5CO \longrightarrow Fe(CO)_5$$

For most other metals, high temperatures and pressures are required to produce the metal carbonyls. By this direct reaction, $Co_2(CO)_8$, $Mo(CO)_6$, $Ru_4(CO)_5$ and $W(CO)_6$ have been prepared when suitable conditions are used.

2. Reductive Carbonylation

This type of reaction involves reducing a metal compound in the presence of CO. The reducing agents may include a variety of materials like Mg, Ag, Cu, Na, H_2, $LiAlH_4$ depending on the particular synthesis being carried out.

$$CrCl_3 + 6CO \xrightarrow[\text{Ether}]{(C_2H_5)_3Al} Cr(CO)_6 + AlCl_3$$

$$MoCl_5 + 6CO \xrightarrow[100^\circ C]{(C_2H_5)_3Al} Mo(CO)_6 + AlCl_3$$

$$OsO_4 + 5CO \xrightarrow[25^\circ C]{350\ \text{atm}} Os(CO)_5 + 2O_2$$

$$2CoI_2 + 8CO + 4Cu \xrightarrow[200\ \text{atm}]{200^\circ C} Co_2(CO)_8 + 4CuI$$

$$FeI_2 + 5CO + 2Cu \xrightarrow[200\ \text{atm}]{200^\circ C} Fe(CO)_5 + Cu_2I_2$$

Yields are often very poor in this process.

Long Question

Q.What is crystal field splitting? Give an account of the important factors which influence the magnitude of crystal field splitting.

Page number 257 onward Haq Nawaz Bhatti

Q.Write optical Isomerism of octahedral complexes?

Page number 323 onward Haq Nawaz Bhatti

Q.How are lanthanides separated from different ores?

Page number 452 onward Haq Nawaz Bhatti

Q.How transuranic actinides were synthesized by artificial transmutation reactions?

Page number 480 onward Haq Nawaz Bhatti

Q.Discuss the John-Teller distortion theorem with examples?

Page number 266 onward Haq Nawaz Bhatti

Q.Write a note on acid opening method of monazite ore for Ln-extraction?

Page number 449 onward Haq Nawaz Bhatti

Beautiful Dua from Quran

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي
وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

Spring

(2024)

Q.What is the role of stability in coordination complexes?

Haq NAwaz Bhatti Page Number 298 onward

OR

https://www.slideshare.net/slideshow/stability-of-co-ordination-complexes-236855156/236855156

Stability of Co-ordination Complexes

The stability of Co-ordination complexes is determined to be its existence in aqueous solution w.r.t its bond dissociation energy, Gibbs free energy (ΔG^0), standard electrode potential (E^0) or pH of the solution . Stability of complexes is also depend on the activation Energy (E_A) or speed of the substitution reaction.

(i) Thermodynamic Stability
(ii) Kinetic Stability

Thermodynamic stability measure in term of complex to exist in equilibrium condition or its measure a tendency of a metal ion to form a particular complex ion.

- → M-L bond energy
- → Stability constant or formation constant (K_f)
- → is directly related to ΔG^0

$$\Delta G^0 \text{ reaction} = \Delta G^0_{product} - \Delta G^0_{reactant}$$
$$\Delta G^0 = -RT\ln K_f$$
$$\Delta G^0 = -nFE^0$$
$$RT\ln K_f = nFE^0$$
$$K_f \propto E^0$$

ΔG° is thermodynamic Property Hence formation Constant is measure as thermodynamic stability. Higher –Ve value of ΔG higher indicate that the complex formation is more stable and spontaneous in nature.

$$[M(aq)]^{X+} + nL \longleftrightarrow [MLn]^{X+}$$

$$K_f = \frac{[MLn]^{X+}}{[M(aq)]^{X+}\ [L]^n}$$

Thermodynamic stability is typically expressed in term of an overall formation or stability constant i.e, K_f

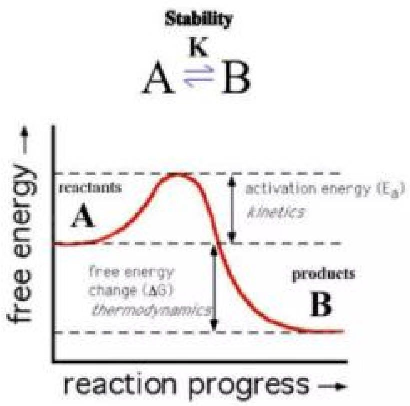
➢The magnitude of formation constant is measure the stability of a complex.

Kinetic Stability

The kinetic stability depends on the speed of the reaction, which depend on activation energy (Ea) and rate of the reaction.

1. Kinetically Inert
2. Kinetically Labile

Inert complex gives substitution reaction with difficulties $t_{1/2} > 1$ min
Labile complexes gives substitution reaction rapidly $t_{1/2} < 1$min



$$[Ni(CN)_4]^{2-} + 4\,{}^{14}CN^{*-} \rightleftharpoons [Ni({}^{14}CN^*)_4]^{2-} + 4CN^-$$

$t_{1/2}$ =30 s **Labile**

$$[Mn(CN)_6]^{3-} + 6\,{}^{14}CN^{*-} \rightleftharpoons [Mn({}^{14}CN^*)_6]^{3-} + 6CN^-$$

$t_{1/2}$ = 1 h **Moderate Labile**

$$[Cr(CN)_6]^{3-} + 6\,{}^{14}CN^{*-} \rightleftharpoons [Cr({}^{14}CN)_6]^{3-} + 6CN^-$$

$t_{1/2}$ = 24 days **Inert**

$$[Cu(NH_3)_4(H_2O)_2]^{2+} + HCL \longrightarrow [CuCl_4]^{2+}$$

Blue Color — Green

Highly Labile

$$[Co(NH_3)_6]^{3+} + \text{Conc. HCl} \longrightarrow \text{No R.K}^n$$

$$[Co(NH_3)_6]^{3+} \xrightarrow[\text{6 M HCl}]{\text{Heat}} [Co(NH_3)_5Cl]^{2+} + NH_4^+$$

Highly Inert

Q.What are metal chelates?

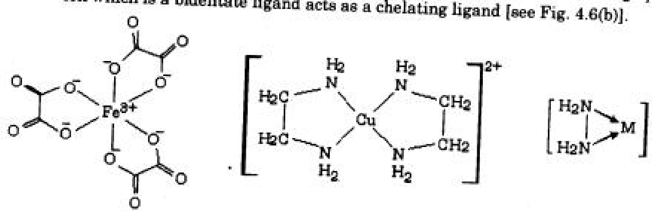
4.7 CHELATING LIGANDS AND CHELATES

A chelating ligand is a bidentate or polydentate ligand which is attached to the same central metal atom by two or more of its donor atoms resulting in the formation of a complex having a strain-free ring structure. The complex having the ring structure is called chelate or chelated complex. The chelate is also called by various other names like *cyclic complex, ring-type complex etc.* The formation of a chelate is called *chelation or cyclisation.*

Examples of the formation of chelates

(i) When two molecules of ethylenediamine, $NH_2 - CH_2 - CH_2 - NH_2$ (en) which is a bidentate ligand, get attached with one Cu^{2+} ion through its two N-donor atoms of each molecule, complex ion viz. $[Cu(en)_2)]^{2+}$ or $[Cu\ (NH_2{-}CH_2{-}CH_2{-}NH_2)_2]^{2+}$ which contains two 5-membered rings is obtained. In the formation of this complex ion, ethylenediamine molecule (bidentate ligand) acts as a chelating ligand and the complex ion thus formed is called a chelate [see Fig. 4.6 (a)].

(ii) When three oxalate ions, $C_2O_4^{2-}$ or ox^{2-} get attached with one Fe^{3+} ion, $[Fe(C_2O_4)_3]^{3-}$ or $[Fe(ox)_3]^{3-}$ ion is obtained. This complex ion is a chelate. This chelate has three 5-membered rings. In the formation of this chelate $C_2O_4^{2-}$ ion which is a bidentate ligand acts as a chelating ligand [see Fig. 4.6(b)].



Q.What is the position of f-block clemens in Periodic table?

[Sanaullah Book Page number 34]

"f-block elements are those elements in which the differentiating electron enters the (n–2)f orbital. These elements are also called inner transition elements and are placed separately at the bottom of the periodic table in two horizontal rows to maintain its structure."

They are part of Group 3 and are divided into two series:

1. Lanthanide Series:
The lanthanide series includes elements with atomic numbers from 58 (Cerium) to 71 (Lutetium). In these elements, the differentiating electron enters the 4f-orbital, and their general electronic configuration is **Xe $4f^{1-14}$ $5d^{0-1}$ $6s^2$**. These elements are commonly known as rare earth elements. Lanthanides mostly exhibit the +3 oxidation state, although some may show **+2 or +4** as well. One of the key features of this series is the lanthanide contraction, which refers to the gradual decrease in atomic and ionic sizes across the series due to poor shielding by 4f electrons. Lanthanides also form coloured ions and are usually paramagnetic, a result of the presence of unpaired electrons in their f-orbitals.

2. Actinide Series:
The actinide series consists of elements with atomic numbers from 90 (Thorium) to 103 (Lawrencium). In these elements, the differentiating electron enters the 5f-orbital, and their general configuration is **Rn $5f^{1-14}$ $6d^{0-1}$ $7s^2$**. Most actinides are radioactive, and several of them are transuranic elements, meaning they come after uranium in the periodic table. Actinides typically exhibit the +3 oxidation state, but they can also show oxidation states from +4 to +7, depending on the element and compound. Similar to the lanthanides, actinides experience actinide contraction, caused by poor shielding of 5f electrons. Because of their radioactivity and ability to release large amounts of energy, actinides are widely used in nuclear reactors and the production of nuclear weapons.

Q.Give names of some Lanthanide ores, from which it can be extreated?

[Repeat in Long Question, Fall 2024]

Q.What are the uses of Uranium other than nuclear fuel?

Although uranium is best known for its use as a nuclear fuel, it also has several other important uses, especially in industry, science, and military. These include:

1. Manufacturing of Armor and Ammunition:

- **Depleted uranium (DU)**, which is less radioactive than natural uranium, is used in the production of armor-piercing bullets and shells.
- It is also used to strengthen military vehicle armor due to its high density and hardness.

2. Use in Aviation and Ship Ballast:

- Because of its very high density, uranium is used as ballast in airplanes, missiles, and ships to help maintain balance and stability.

3. Production of Colored Glass and Ceramics:

- Uranium compounds (like uranium oxide) were historically used to make uranium glass (which has a greenish-yellow color and glows under UV light).
- It is also used as a coloring agent in ceramics and glazes.

4. Scientific Research and Radiography:

- Uranium is used in radiographic testing to check for cracks and flaws in metal parts (non-destructive testing).
- In science, it is used as a target material in particle accelerators and source material for the production of other radioactive isotopes.

5. Dating of Ancient Rocks (Uranium-Lead Dating):

Uranium isotopes are used in radiometric dating techniques to estimate the age of rocks, fossils, and **Earth's crust** (Uranium-238 decays to lead-206).

Q.Explain the structure of Hexaammine Iron (11) chloride and Hexazammine copper (b) by VBT?

Q.What is the commonly found actinides in earth crust?

Haq Nawaz Bhaati page number 480 onward

Commonly Found Actinides in Earth's Crust

The actinide series includes elements from atomic numbers 89 to 103, but only four actinides occur naturally in the Earth's crust: Thorium (Th), Uranium (U), Protactinium (Pa), and Actinium (Ac). The rest are artificially produced in nuclear reactors. These naturally occurring actinides are important due to their availability and applications in energy, defense, and research.

1. Thorium (Th)
Thorium (atomic number 90) is the most abundant natural actinide. It is commonly found in monazite sands, especially in regions like India and Brazil. It exists mainly as thorium oxide (ThO_2). Thorium is a fertile nuclear material, meaning it can be converted into fissile uranium-233 inside reactors, making it valuable for nuclear power in the future.

Applications of Thorium:

- ❖ Thorium is used in thorium-based nuclear reactors as an alternative to uranium fuel.
- ❖ It is used in gas mantles for portable lighting because thorium oxide glows brightly when heated.
- ❖ It is also used in high-temperature alloys for aerospace and industrial applications due to its strength and heat resistance.

2. Uranium (U)
Uranium (atomic number 92) is the best-known actinide and occurs in minerals like pitchblende (UO_2) and carnotite. It naturally exists as two main isotopes: U-238 (99.3%) and U-235 (0.7%). U-238 is used in breeder reactors, while U-235 is a fissile material widely used in nuclear power and weapons.

Applications of Uranium:

- ❖ Uranium is the primary fuel used in nuclear power plants for electricity generation.
- ❖ It is used in the production of nuclear weapons and military armor, especially depleted uranium (DU).
- ❖ Uranium isotopes are used in radiometric dating, especially U-Pb dating, to determine the age of rocks and Earth's crust.

3. Protactinium (Pa)
Protactinium (atomic number 91) is a rare and highly radioactive element. It is found in very small quantities in uranium ores as a decay product of uranium-235. Due to its scarcity and strong radioactivity, it has no significant industrial applications but is of academic interest.

Applications of Protactinium:

- ❖ Protactinium is used mainly in scientific and nuclear research to study radioactive decay chains and actinide behavior.

4. Actinium (Ac)
Actinium (atomic number 89) is present in trace amounts in uranium ores. It is a soft, silvery radioactive metal and forms as a decay product of uranium. Though rare, it is very useful in certain nuclear and medical applications due to its strong alpha radiation.

Applications of Actinium:

- ❖ Actinium-225 is used in targeted alpha therapy (TAT) for the treatment of cancer.
- ❖ Actinium is also used in neutron sources when combined with beryllium (Ac-Be) for research and scientific instruments.

Q.Give some examples of Actinides produced by artificial radioactivity?

Haq Nawaz bhatti page number 484

Artificial radioactivity allows scientists to create actinides that do not naturally exist in significant amounts. These transuranic elements (with atomic numbers greater than 92) are mainly produced by bombarding uranium or plutonium with neutrons or alpha particles in nuclear reactors or particle accelerators.

Actinides produced by artificial radioactivity are typically transuranic elements synthesized through nuclear reactions like neutron capture or heavy ion bombardment. Examples include:

- **Neptunium (Np, Z=93):** Formed by neutron capture on uranium-238, followed by beta decay:

$$^{238}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{239}_{92}\mathrm{U} \xrightarrow{\beta-} ^{239}_{93}\mathrm{Np}$$

- **Plutonium (Pu, Z=94):** Produced from neptunium-239 via beta decay:

$$^{239}_{93}\mathrm{Np} \xrightarrow{\beta-} ^{239}_{94}\mathrm{Pu}$$

- **Americium (Am, Z=95):** Synthesized by neutron capture on plutonium-239, followed by beta decay:

$$^{239}_{94}\mathrm{Pu} + 4n \rightarrow ^{243}_{94}\mathrm{Pu} \xrightarrow{\beta-} ^{243}_{95}\mathrm{Am}$$

- **Curium (Cm, Z=96):** Formed by neutron capture on americium-243, followed by beta decay:

$$^{243}_{95}\mathrm{Am} + n \rightarrow ^{244}_{95}\mathrm{Am} \xrightarrow{\beta-} ^{244}_{96}\mathrm{Cm}$$

- **Lawrencium (Lr, Z=103):** Produced via heavy ion bombardment, for example:

$$^{249}_{98}\mathrm{Cf} + ^{11}_{5}\mathrm{B} \rightarrow ^{256}_{103}\mathrm{Lr} + 4^{1}_{0}\mathrm{n}$$

Q.Which of the complex ion namely $[Fe(CN)_6]^{3-}$ and $[Fe(H_2O)_6]^{3+}$is an outer orbital complex and why?

$[Fe(CN)_6]^{3-}$ – Inner Orbital Complex

In this complex, iron is in the +3 oxidation state **(Fe^{3+})**, which gives it a $3d^5$ electronic configuration. The ligand CN^- is a strong field ligand, which causes the pairing of electrons in the 3d orbitals. Due to this pairing, two 3d orbitals become vacant and are available for bonding. These two inner 3d orbitals, along with one 4s and three 4p orbitals, participate in hybridization to form **d^2sp^3** hybridization. Since the complex uses inner **(n–1)d** orbitals, it is classified as an inner orbital complex. Because of electron pairing, the complex is low-spin, and it contains only one unpaired electron, making it Weakly paramagnetic.

$[Fe(H_2O)_6]^{3+}$ – Outer Orbital Complex

In this complex, iron is also in the +3 oxidation state**(Fe^{3+})** with a $3d^5$ configuration. The ligand H_2O is a weak field ligand, which does not cause pairing of the 3d electrons. As a result, all five 3d orbitals remain singly occupied and are not available for bonding. Therefore, the complex uses outer orbitals—one 4s, three 4p, and two 4d orbitals—for bonding, resulting in **sp^3d^2** hybridization. Since outer (n)d orbitals are involved, it is called an outer orbital complex. Because no electron pairing occurs, the complex is high-spin with five unpaired electrons, making it strongly paramagnetic.

Structure and electronic configuration available sanullanh book page number 205 and 205

Q.What is the difference in optical and geometrical isomerism?

Repeat in long Fall 2024

Q.Give two applications of Lanthanide alloys?

Haq Nawaz bhatti page number 465

1. Permanent Magnets (NdFeB Alloys)

Lanthanide alloys play a crucial role in modern technology, particularly in the production of high-strength neodymium-iron-boron (NdFeB) magnets. These magnets are the strongest permanent magnets available, offering exceptional magnetic properties that make them indispensable in various applications. They are widely used in electric vehicle motors, where they improve efficiency and power output, and in wind turbine generators, enabling renewable energy production. Additionally, NdFeB magnets are found in everyday electronics like headphones, hard disk drives, and MRI machines, showcasing their versatility and importance in both industrial and consumer sectors.

2. Pyrophoric and Structural Alloys (Mischmetal)

Another significant application of lanthanide alloys is mischmetal, a blend of cerium, lanthanum, and other light lanthanides with iron. This alloy is highly valued for its pyrophoric properties, making it ideal for flints in lighters and ignition devices, where it produces sparks upon friction. Beyond pyrophoric uses, mischmetal is added to magnesium and steel alloys to enhance their strength, durability, and workability. For instance, it is used in military applications like bullets and shells, as well as in aerospace and automotive components, where lightweight yet robust materials are essential.

Q.How MOT is superior to VBT?

VBT	MOT
1. According to VB approach only valence electrons are involved in bond formation.	1. According to MO approach all the electrons of interacting atoms are involved in bond formation.
2. In VBT, the two concerned atoms do not lose their individual identity.	2. In MOT, the two atoms lose their individual identity.
3. It does not offer a simple method of excited states in molecules.	3. It offers a simple and more convenient description of excited states of molecules
4. Resonance plays an important role in VBT.	4. Resonance plays no role in MOT.
5. It does not explain the paramagnetic behaviour of molecules like O_2 molecule.	5. It explains the paramagnetic behaviour of molecules like O_2 molecule

Haq Nawaz bhatti page number 194

Q.How magnetism is explained by CFT?

Haq Nawaz bhatti page number 279

Equation (iv) shows that the magnitude of μ depends on the number of unpaired electrons (n) present in a given species. The value of μ for different number of unpaired electrons can be calculated with the help of equation (iv) as shown below:

For $n = 0$ $\mu_s = \sqrt{0\,(0+2)} = 0.0$ B.M.

For $n = 1$ $\mu_s = \sqrt{1\,(1+2)} = \sqrt{3} = 1.73$ B.M.

5.7.9 Crystal Field Theory (CFT) and Magnetic Properties of Complexes:

CFT has helped us to understand the magnetic properties of coordination compounds. The substances which have all paired electrons are called diamagnetic. The substances which contain unpaired electrons are called paramagnetic substances. Magnitude of Δ_0 and pairing energy P play an important role in this direction.

(i) If $\Delta_0 > P$, the electrons will prefer to pair up and the complex will be **low spin complex**.

(ii) If $P > \Delta_0$, the electrons will not pair up and the complex will be **high spin complex**.

Thus, the complexes with **weak field ligands** are **high spin complexes and** are paramagnetic. Those with **strong field ligand** are **low spin complexes** are diamagnetidc.

This generalization could not be provided by valence bond theory. Consider the following two complexes of cobalt (III). From experiments, it has been observed that the complex $[Co(NH_3)_6]^{3+}$ is diamagnetic while the complex $[CoF_6]^{3-}$ is **paramagnetic**.

Cobalt in $\mathbf{[Co(NH_3)_6]^{3+}}$ and $\mathbf{[CoF_6]^{3-}}$ is in **+3** oxidation state and it has $\mathbf{d^6}$ configuration Fig. (5.32). It has been observed that $F^{\ominus}$ ion is a weak field ligand and therefore Δ_0 is small. Therefore, Δ_0 will be less than P and the electron will remain unpaired as far as possible. This is obviously (صاف اندازے) because greater energy is required in pairing and smaller energy is required in moving to orbital of higher energy.

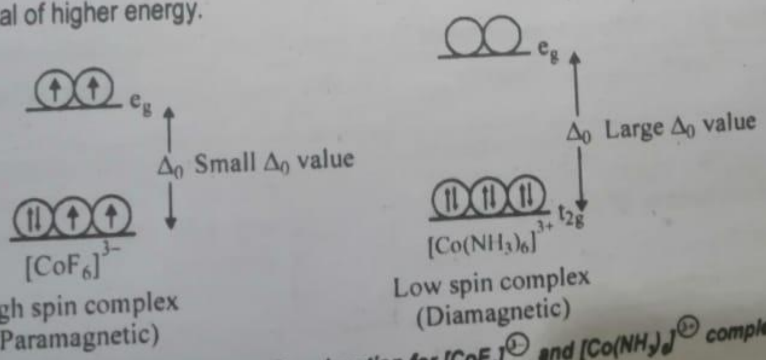


Fig. (5.32): Crystal field splitting and explanation for $[CoF_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$ complexes.

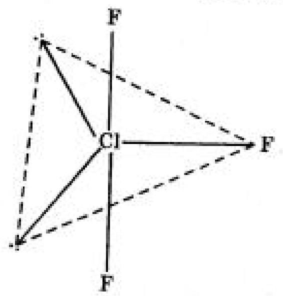
R. What is the shape of AB3E2, molecule according to VSEPR theory?

[Haq Nawaz bhatti page number 141](#)

...etry of SF_4 molecule

9. AB_3E_2 Type

Here A is the central atom, B are the bonded atoms to the central atom and E are the lone pairs of electrons around the central atom. Species of this type have three bonding electron pair and two lone pairs of electrons (bp = 3, lp = 2). These bonding electron pairs and lone pairs of electrons occupy their positions as far apart as possible so that the electrostatic repulsions between them is minimum. This is most probable if the species of this type has T-shaped bent geometry in which both the lone pairs occupy the equatorial positions of the trigonal bipyramidal geometry. This position of lone pairs involves minimum repulsions as compared to axial position. The bond angle between two axial B-atoms is less than 180° and between equatorial and axial atoms is less than 90°. Example, ClF_3, BrF_3 $1F_3$, ICl_3 etc. All these molecules have T-shaped geometry.



$_{17}Cl = 1s^2, 2s^2, 2p^6, 3s^2, 3p_x^2, 3p_y^2, 3p_z^1$

$_{17}Cl^* = 1s^2, 2s^2p^6, 3s^2, 3p_x^2, 3p_y^1, 3p_z^1, 3d^1$

Fig.3.9: T-shaped geometry of ClF_3 molecule

Q.What is the formula to calculate stability constant?

The stability constant quantifies the equilibrium between a **metal ion (M)** and **ligands (L)** forming a **complex ion** (ML_n) in solution:

$$K_{\text{stab}} = \frac{[\mathrm{ML}_n]}{[\mathrm{M}][\mathrm{L}]^n}$$

Where:

- $[\mathrm{ML}_n]$ = Equilibrium concentration of the complex ion.
- $[\mathrm{M}]$ = Equilibrium concentration of the free metal ion.
- $[\mathrm{L}]$ = Equilibrium concentration of the free ligand.
- n = Number of ligands bound to the metal ion.

Example: $[Cu(NH_3)_4]^{2+}$ Formation

For the reaction:

$$[\mathrm{Cu(H_2O)_6}]^{2+} + 4\,\mathrm{NH_3} \rightleftharpoons [\mathrm{Cu(NH_3)_4(H_2O)_2}]^{2+} + 4\,\mathrm{H_2O}$$

The stability constant is:

$$K_{\text{stab}} = \frac{[\mathrm{Cu(NH_3)_4(H_2O)_2}]^{2+}}{[\mathrm{Cu(H_2O)_6}]^{2+}[\mathrm{NH_3}]^4}$$

Beautiful Dua from Quran

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي

وَاحْلُلْ عُقْدَةً مِّن لِّسَانِي يَفْقَهُوا قَوْلِي

Punjab University

Bs (4 Years) Chemistry

Semester Sixth

Solve Past paper

Fall (2023)

Q.What are the structural theories of coordination complexes?

The structural theories of coordination complexes explain the bonding, geometry, and properties of these compounds. The main theories are:

1. Werner's Theory

Werner's Theory of Coordination Compounds defines coordination compounds as entities where the central metal atom exhibits two types of valencies: primary valency and secondary valency. The primary valency corresponds to the oxidation state of the metal and is ionizable, satisfied by negative ions, and written outside the coordination sphere. The secondary valency corresponds to the coordination number, is non-ionizable, satisfied by ligands (neutral molecules or negative ions), and written inside the coordination sphere. Werner's theory explains that the spatial arrangement of ligands around the metal ion is fixed and gives the complex a definite geometry, such as octahedral or tetrahedral. This theory was the first to successfully explain the structures and isomerism of coordination compounds by distinguishing between these two valencies, although it could not explain properties like color or magnetism

2. Valence Bond Theory (VBT)

Valence Bond Theory (VBT) defines the bonding in coordination complexes as the formation of coordinate covalent bonds through the overlap of ligand lone pair orbitals with vacant hybrid orbitals of the central metal ion. According to VBT, the metal ion undergoes hybridization **(such as d2sp3, sp3d, or sp3)** to accommodate ligands, which determines the geometry of the complex. This theory explains the magnetic properties and geometries of complexes by considering the involvement of inner or outer d orbitals, distinguishing between low-spin and high-spin complexes. VBT provides a qualitative understanding of bonding and geometry but does not fully account for electronic spectra or the nature of metal-ligand bonding beyond covalent interactions.

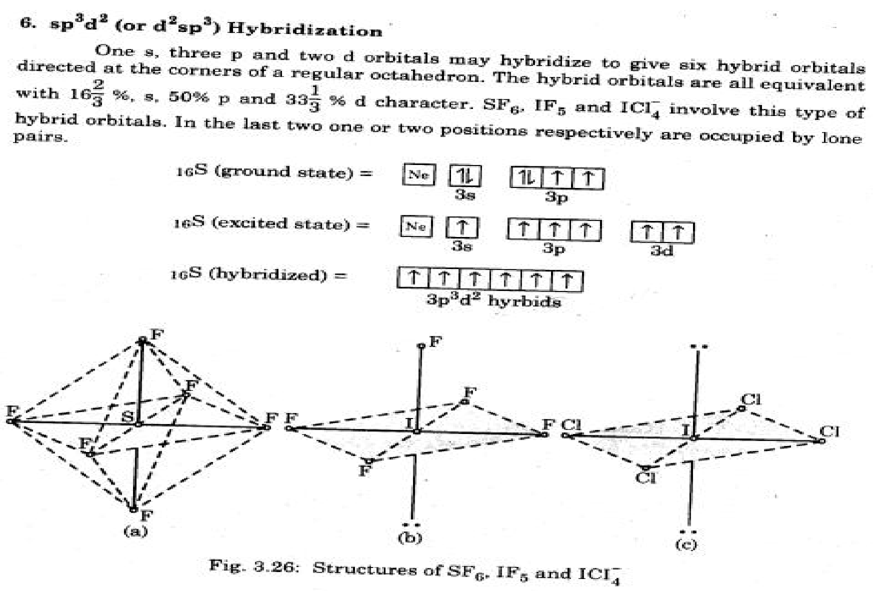
3. Crystal Field Theory (CFT)

Crystal Field Theory (CFT) defines the interaction between the metal ion and ligands as purely electrostatic, where ligands are treated as point charges that cause the splitting of the metal ion's d orbitals into groups of different energies. This splitting explains the color, magnetic behavior, and stability of coordination complexes. CFT predicts the geometry of complexes (octahedral, tetrahedral, square planar) based on the pattern of d orbital splitting and the ligand field strength. It provides a quantitative approach to understanding electronic transitions and magnetic properties but neglects covalent bonding aspects.

4. Ligand Field Theory (LFT) / Molecular Orbital Theory (MOT)

Ligand Field Theory (LFT), an extension of CFT, defines the bonding in coordination complexes by incorporating molecular orbital theory to describe the covalent character of metal-ligand bonds. LFT considers the mixing of metal d orbitals with ligand orbitals, providing a more detailed and accurate description of electronic structure, bonding, and magnetic properties. It explains phenomena that CFT and VBT cannot fully address, such as the nature of bonding orbitals and the influence of ligand symmetry on complex properties.

Q.What is hybridization in AB6 molecules?



6. sp^3d^2 (or d^2sp^3) Hybridization

One s, three p and two d orbitals may hybridize to give six hybrid orbitals directed at the corners of a regular octahedron. The hybrid orbitals are all equivalent with $16\frac{2}{3}$ %, s, 50% p and $33\frac{1}{3}$ % d character. SF_6, IF_5 and ICl_4^- involve this type of hybrid orbitals. In the last two one or two positions respectively are occupied by lone pairs.

$_{16}S$ (ground state) = [Ne] ⇅ (3s) ⇅ ↑ ↑ (3p)

$_{16}S$ (excited state) = [Ne] ↑ (3s) ↑ ↑ ↑ (3p) ↑ ↑ (3d)

$_{16}S$ (hybridized) = ↑ ↑ ↑ ↑ ↑ ↑ ($3p^3d^2$ hyrbids)

Fig. 3.26: Structures of SF_6, IF_5 and ICl_4^-

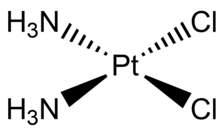
Q.Which theory explains paramagnetism of oxygen?

The theory that successfully explains the paramagnetism of **oxygen (O_2) is Molecular Orbital Theory (MOT).**

While Valence Bond Theory (VBT) and Lewis structures fail to account for oxygen's observed paramagnetism (they predict all electrons to be paired, making it diamagnetic), MOT accurately predicts that the oxygen molecule has two unpaired electrons in its antibonding **π* (pi-star) (π2py* and π2pz*)**molecular orbitals. These unpaired electrons give the oxygen molecule a net magnetic moment, causing it to be attracted to a magnetic field, thus exhibiting paramagnetism.

[Structure for VBT and MOT Repeat]

Q.Which metal complex is used in cancer therapy?



(cis-diamminedichloridoplatinum(II))

The most prominent and widely used metal complexes in cancer therapy are platinum-based drugs, with cisplatin **(cis-diamminedichloridoplatinum(II)) $[Pt(NH_3)_2Cl_2]$** being the pioneering and most famous example. Discovered in the late **1960s** and approved by the FDA in **1978**, cisplatin is highly effective against various solid tumors, including ovarian, testicular, bladder, lung, head and neck cancers, and some lymphomas. Its anticancer activity arises from forming DNA cross-links that disrupt replication and transcription, leading to cancer cell death (apoptosis).

Due to cisplatin's significant side effects such as nephrotoxicity, neurotoxicity, and ototoxicity, other platinum drugs like **carboplatin** and **oxaliplatin** were developed. Carboplatin has a different toxicity profile with less kidney damage but more bone marrow suppression, while oxaliplatin is mainly used for colorectal cancer and causes peripheral neuropathy.

Q.Why are f-block elements are called rare earth elements?

The f-block elements, particularly the lanthanides **(atomic numbers 57-71)**, are commonly referred to as rare earth elements due to a combination of historical and geological factors. When these elements were first discovered in the late **18th and 19th centuries,** they were initially thought to be extremely scarce in nature, hence the term **"rare."** However, we now know that many of these elements are actually relatively abundant in Earth's crust - some, like cerium, are nearly as common as copper. The "earth" part of their name comes from early chemists who classified metal oxides as **"earths,"** as these elements were typically isolated from mineral oxides like **yttria and ceria**.

These elements are called **"rare"** because they are spread out in small amounts in many different minerals, not found in large, concentrated deposits. Also, they are very similar to each other chemically, which makes it hard and expensive to separate them from each other. Because of these difficulties, they kept the name "rare earth elements" even after scientists realized they are not actually very rare. Today, these elements are very important for many modern technologies like smartphones, electric cars, wind turbines, and military equipment.

Haq Nawaz Bhatti Page Number 440

Q.How are coordination complexes prepared?

Haq Nawaz Bhatti Page Number 310 to 314 , you can write according your need

Q.How lanthanides can be separated from actinides?

Repeat in Fall 2024

Q. What is the type of hybridization in $Fe(CO)_5$ and $[Ni(CN)_4]^{2-}$?

4. dsp^2 Hybridization

dsp^2 hybridization involves the mixing of one s, two p and one d orbitals on the same atom to form four identical dsp^2 hybrids. Each four hybrid orbitals are directed to the corners of a square giving a square planar structure. The formation of $[Ni(CN)_4]^{2-}$ ion can explained on dsp^2 hybridization. This molecule has square planar geometry as shown in figure 3.22. This hybridization is mostly encountered in coordination compounds.

$$_{28}Ni = 1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2, 3d^8$$

$$_{28}Ni^{2+} = 1s^2, 2s^2 2p^6, 3s^2 3p^6, 3d^8$$

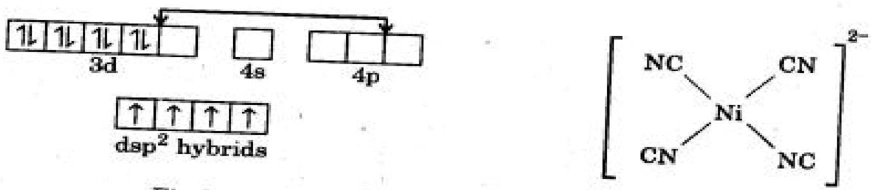


Fig.3.22: Formation of $[Ni(CN)_4]^{2-}$ ion.

$Fe(CO)_5$

Available Sanaullah book page number 203 onward , Same concept above mention

(a) How does the coordination number of a metal affect the stability of a complex?

Haq Nawaz Bhatti Page Number 306 to 310, you can write according your need

Q.How are the colors of coordination compounds explained by CFT?

Haq Nawaz Bhatti Page Number 276

OR

The colors of coordination compounds are explained by Crystal Field Theory through the splitting of the metal ion's d-orbitals in the presence of ligands and the resulting electronic transitions known as d-d transitions. When ligands approach a metal ion, the five degenerate d-orbitals split into different energy levels depending on the geometry of the complex. **For example,** in an octahedral complex, the d-orbitals split into a lower-energy set called **t2g** and a higher-energy set called **eg**, separated by an energy gap called the crystal field splitting energy **Δ0.**
When white light shines on the complex, electrons in the lower-energy t2g orbitals absorb light energy equal to Δ0 and get excited to the higher-energy egeg orbitals. The wavelength of light absorbed corresponds to this energy difference. The color we observe is the complementary color of the absorbed light because the unabsorbed wavelengths are transmitted or reflected.

Several factors influence the color of coordination compounds:

Oxidation state of the metal:
Higher oxidation states increase Δ0, shifting absorption to shorter wavelengths (blue shift).

Nature of ligands:
Strong-field ligands **(like CN^-or CO)** cause larger splitting and absorption of higher-energy light, while weak-field ligands **(like H_2O or Cl^-)** cause smaller splitting.

Coordination geometry:
Different geometries (octahedral, tetrahedral, square planar) produce different splitting patterns and magnitudes, affecting color.

Some complexes with **d^0 or d^{10}** configurations do not show color because they lack d-d transitions. Additionally, some colors arise from charge transfer transitions rather than d-d transitions.

Example:
The complex $[Ti(H_2O)_6]^{3+}$ absorbs green light (~500 nm), so it appears purple, the complementary color of green.

CFT explains the color of coordination compounds by the splitting of d-orbitals in the ligand field, absorption of specific wavelengths of light corresponding to this splitting, and the transmission of the complementary color.

Q.What is the difference between a hard and a soft ligand in coordination chemistry?

Sanaullah Page Number 138 to 139 Answer in symbiosis paragraph

OR

In coordination chemistry, ligands are classified as hard or soft based on their electronic and structural properties, which influence their bonding preferences with metal ions.

Hard ligands

Hard ligands are typically small, have high electronegativity, and low polarizability. They usually contain donor atoms like oxygen, nitrogen, or fluorine that hold their electrons tightly. Hard ligands

tend to form ionic or electrostatic bonds with metal ions, preferring to bind to hard metal ions (which are small with high positive charge).
Examples of hard ligands include F^-, OH^-, NH3, and H_2O.

Soft ligands

Soft ligands, on the other hand, are larger, more polarizable, and have lower electronegativity. They contain donor atoms such as sulfur, phosphorus, or iodine, with electrons more easily distorted. Soft ligands tend to form covalent bonds with metal ions and prefer to bind to soft metal ions (which are larger with lower positive charge and often have filled or partially filled d orbitals).
Examples of soft ligands include I^-, S^{2-}, PR_3 (phosphines), and CN^-.

This classification is part of the Hard and Soft Acids and Bases (HSAB) theory, which states that hard acids prefer to bind to hard bases (hard ligands), and soft acids prefer to bind to soft bases (soft ligands), resulting in more stable complexes.

Q.What is the difference between square planar and tetrahedral complexes?

https://chem.libretexts.org/Courses/Douglas_College/DC%3A_Chem_2330_(O'Connor)/4%3A_Crystal_Field_Theory/4.3%3A_High_Spin_and_Low_Spin_Complexes

OR

https://www.differencebetween.com/difference-between-square-planar-and-tetrahedral-complexes/

Aspect	Square Planar Complexes	Tetrahedral Complexes
Geometry	The four ligands are arranged at the corners of a square in the same plane around the central metal ion.	The four ligands are positioned at the corners of a tetrahedron around the central metal ion.
Bond Angles	The bond angles between adjacent ligands are **90°**, and trans ligands are **180°** apart.	The bond angles between all ligands are approximately **109.5°**.
Coordination Number	The coordination number is 4.	The coordination number is also 4.
Common Metal Electron Config.	Most common for **d^8** metals (e.g., Ni^{2+}, Pd^{2+}, Pt^{2+}, Au^{3+}) with **strong-field ligands**.	Common for **d^0, d^{10}, or weak-field d^7/d^9** metals (e.g., Zn^{2+}, Cd^{2+}, Co^{2+}, Cu^+).
Crystal Field Splitting	The d-orbitals split into **four levels**: $d_{x^2-y^2}$ (highest energy) > d_{z^2} > d_{xy} > d_{xz}/d_{yz}.	The d-orbitals split into **two levels**: t_2 (dxy, dxz, dyz) > e (dz^2, dx^2-y^2). **Splitting energy (Δ_t) is smaller (≈4/9 Δ_o)**.

Magnetic Properties	Usually **low-spin/diamagnetic** (e.g., $[Ni(CN)_4]^{2-}$).	Often **high-spin/paramagnetic** (e.g., $[CoCl_4]^{2-}$), except d^{10} (e.g., $[ZnCl_4]^{2-}$).
Examples	$[Pt(NH_3)_2Cl_2]$ (cisplatin), $[Ni(CN)_4]^{2-}$, $[PdCl_4]^{2-}$.	$[ZnCl_4]^{2-}$, $[MnCl_4]^{2-}$, $[NiCl_4]^{2-}$ (high-spin Ni^{2+}).

Q.How the stability of a coordination compound can be checked?

Repeat

Q.How do coordination compounds play a role in biological systems?

Repeat in Fall 2024

Q.What is the difference between high spin and low spin complex in coordination chemistry?

Haq Nawaz Bhatti Page Number 270 Onward

Aspect	**High Spin Complex**	**Low Spin Complex**
Ligand Type	High spin complexes are formed when the ligands are weak field ligands, such as water (H_2O), fluoride (F^-), or chloride (Cl^-).	Low spin complexes are formed when the ligands are strong field ligands, such as cyanide (CN^-), carbonyl (CO), or ammonia (NH_3).
Crystal Field Splitting	In high spin complexes, the crystal field splitting energy (Δ) is small, so electrons can occupy higher orbitals easily.	In low spin complexes, the crystal field splitting energy (Δ) is large, so electrons prefer to pair up in lower orbitals.
Electron Configuration	Electrons tend to occupy all orbitals singly first before pairing (Hund's rule).	Due to large splitting, electrons in low spin complexes prefer to pair in the lower energy orbitals to avoid high energy states.
Number of Unpaired Electrons	High spin complexes usually have a greater number of unpaired electrons, depending on the metal and its oxidation state.	Low spin complexes usually have fewer or no unpaired electrons, leading to different magnetic properties.
Magnetic Properties	High spin complexes are generally paramagnetic because of the presence of unpaired electrons.	Low spin complexes are often diamagnetic or weakly paramagnetic, due to electron pairing.
Examples	An example of a high spin complex is $\mathbf{[Fe(H_2O)_6]^{3+}}$, where water is a weak field ligand and more electrons remain unpaired.	An example of a low spin complex is $\mathbf{[Fe(CN)_6]^{3-}}$, where cyanide is a strong field ligand and electrons pair up in lower orbitals.